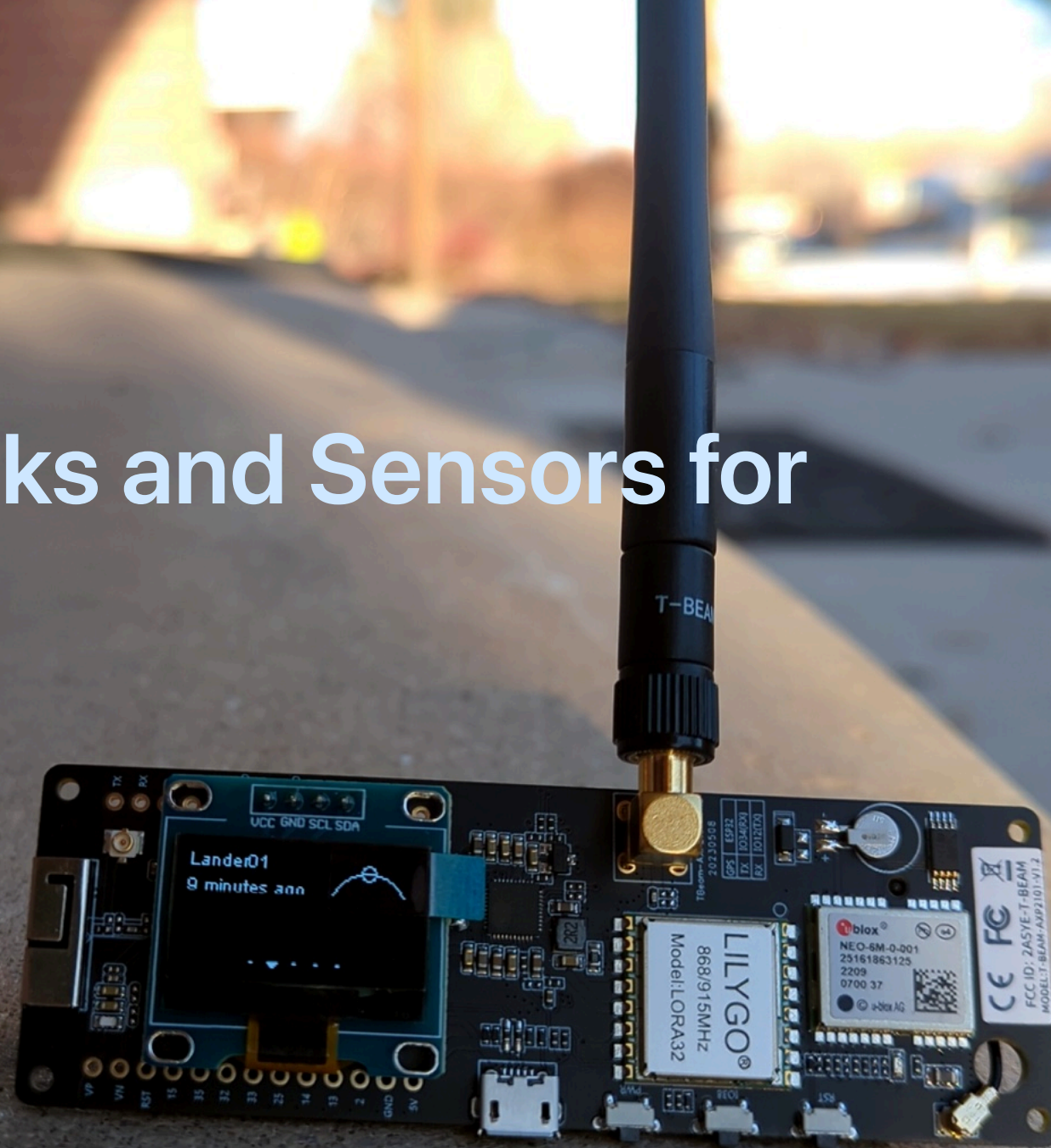


Mesh Networks and Sensors for Communities

Jake Hosen



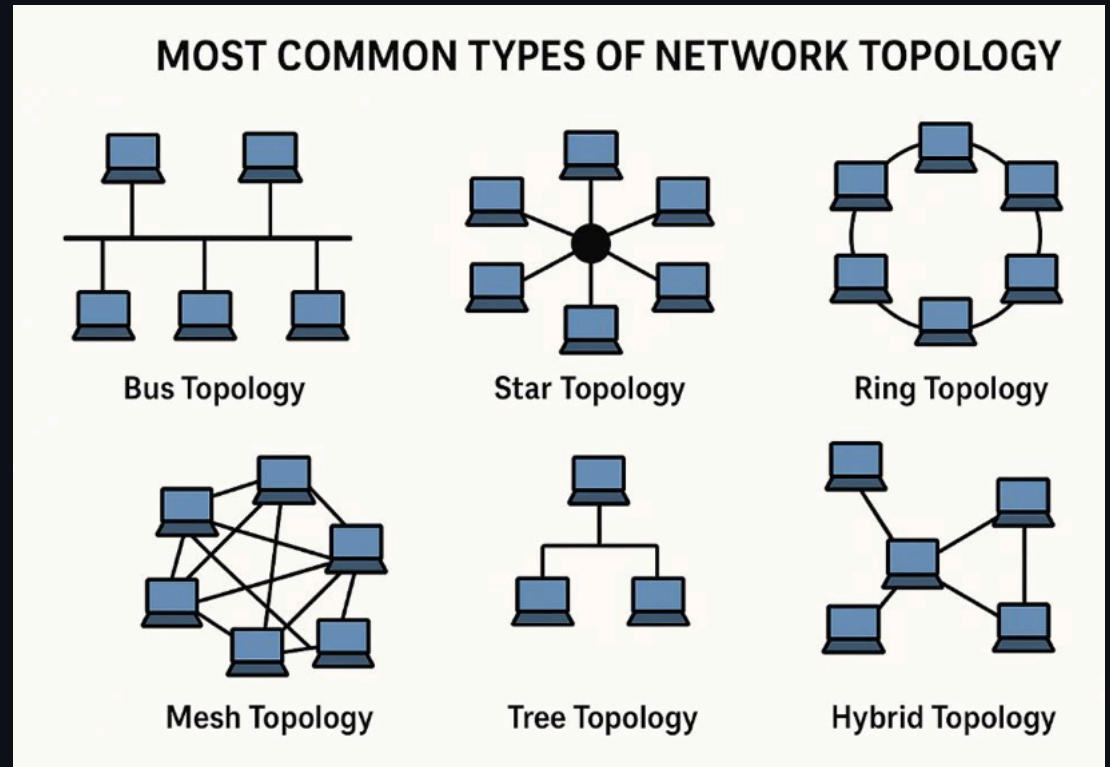
Two Topics

1. **Mesh Networks: what are they and how could we use them for communication?**
2. **Sensor Networks: Communities can use sensors for empowerment.**

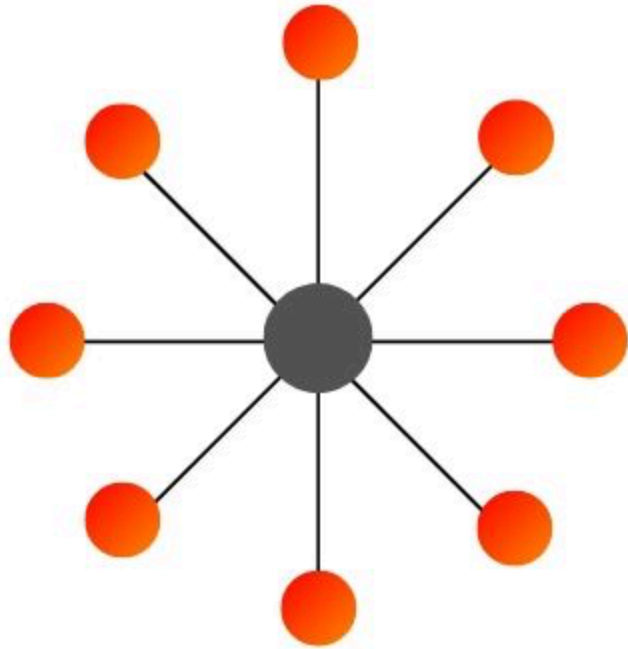
Mesh Networks

A network layout (topology) that maximizes connectivity and minimizes the chances of a network failure.

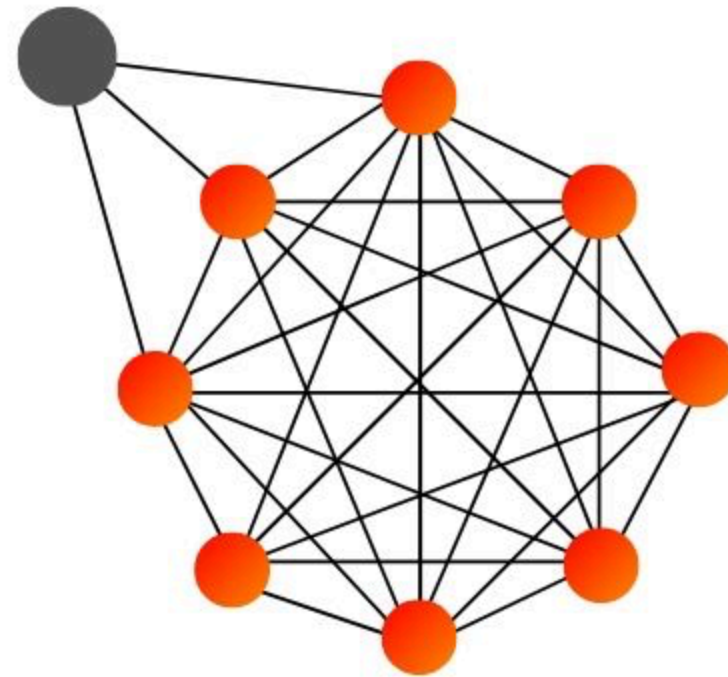
Developing such stable systems was not easy.



Mesh Topology = More Redundancy



Star Topology

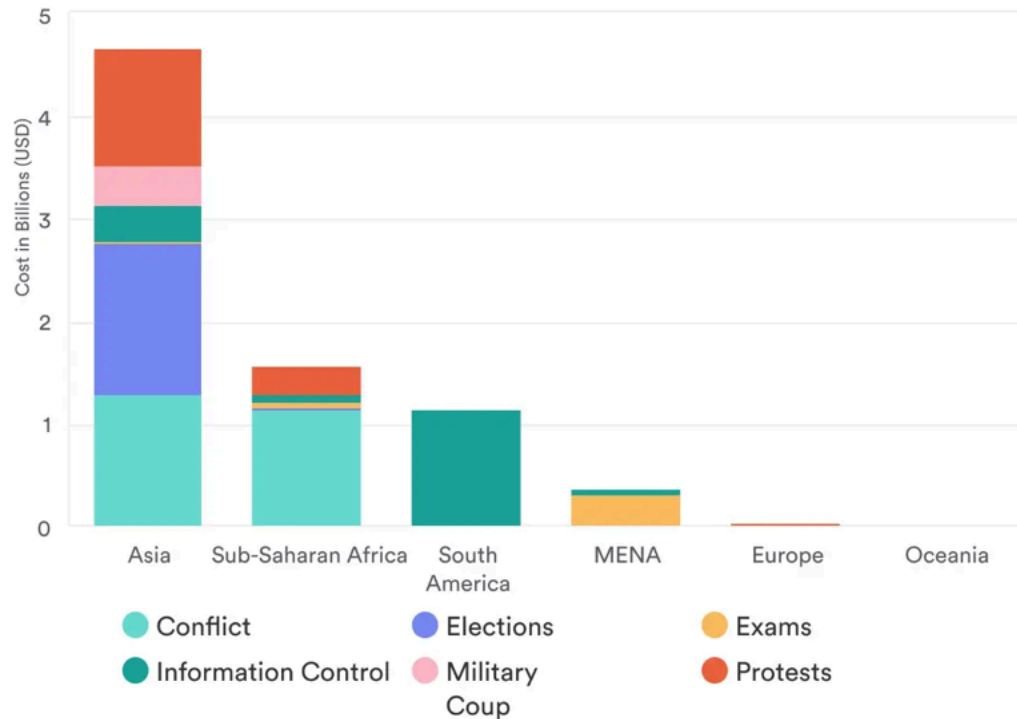


Mesh Topology

BOOST
HIGH

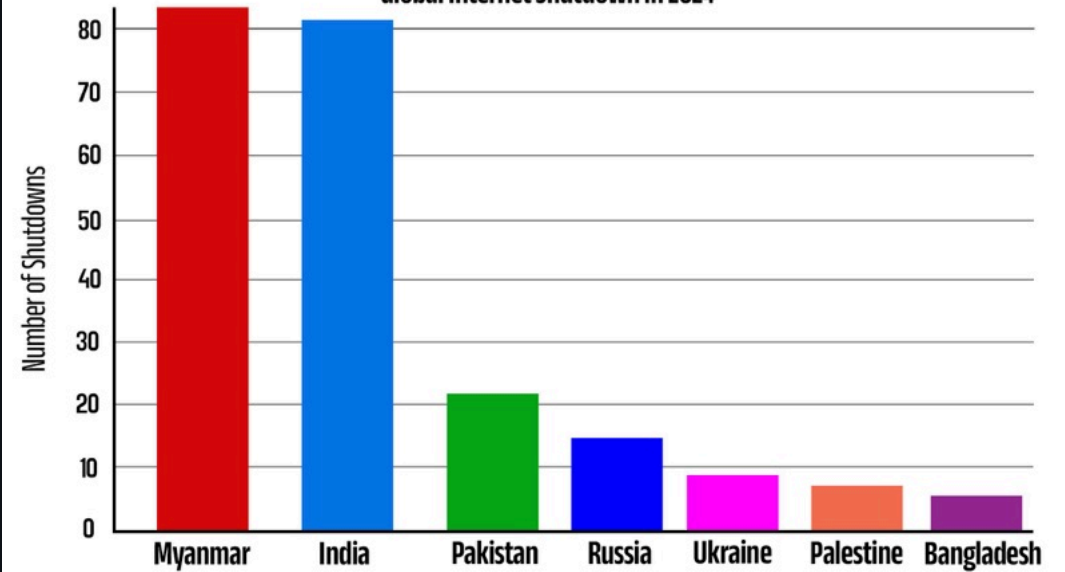
Why Do We Care

Why Did Governments Around The World Block the Internet in 2024?



Research by Top10VPN 10

Global Internet Shutdown in 2024



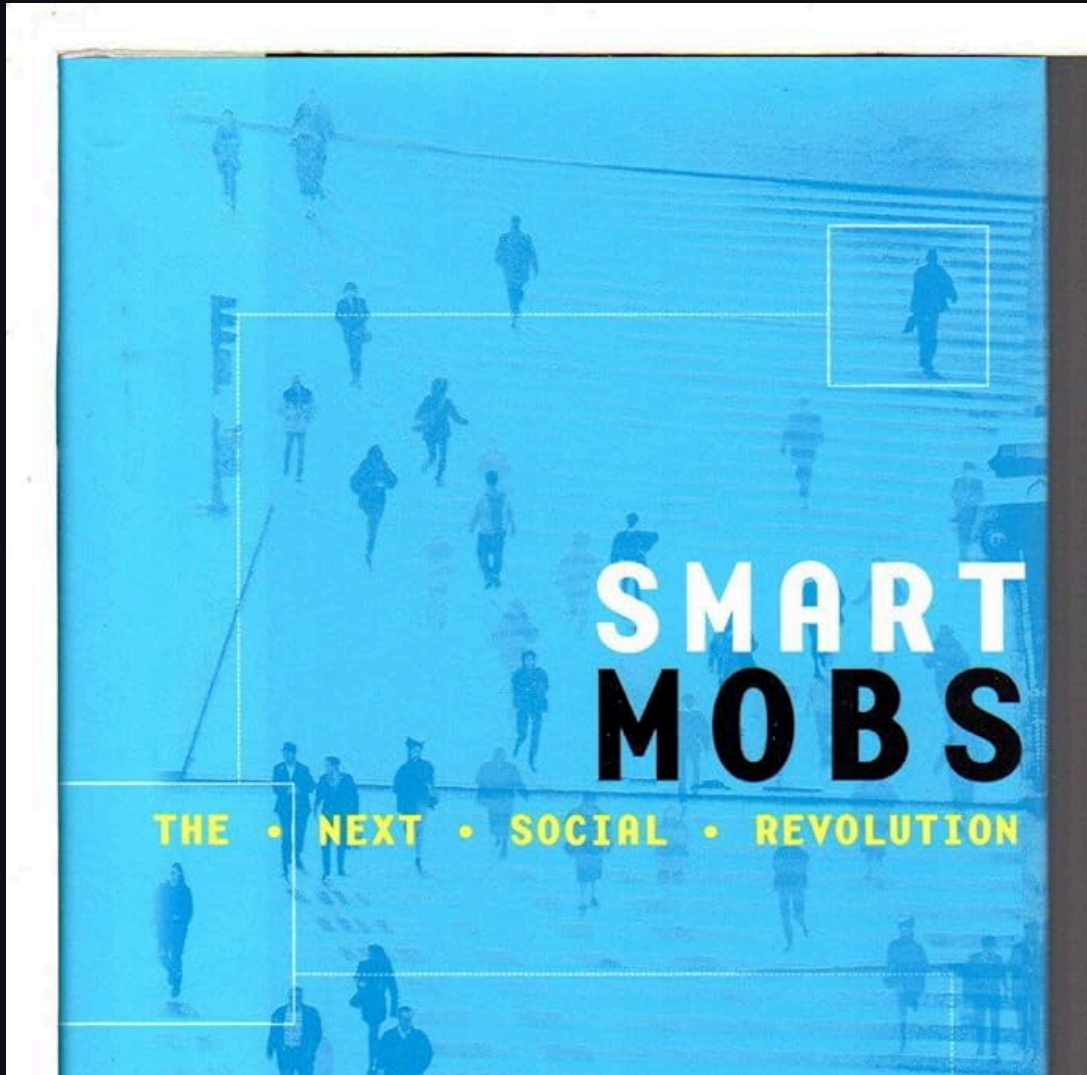
In the US

No internet "kill switch". Congress has tried to give the president this power including under Obama in 2011.

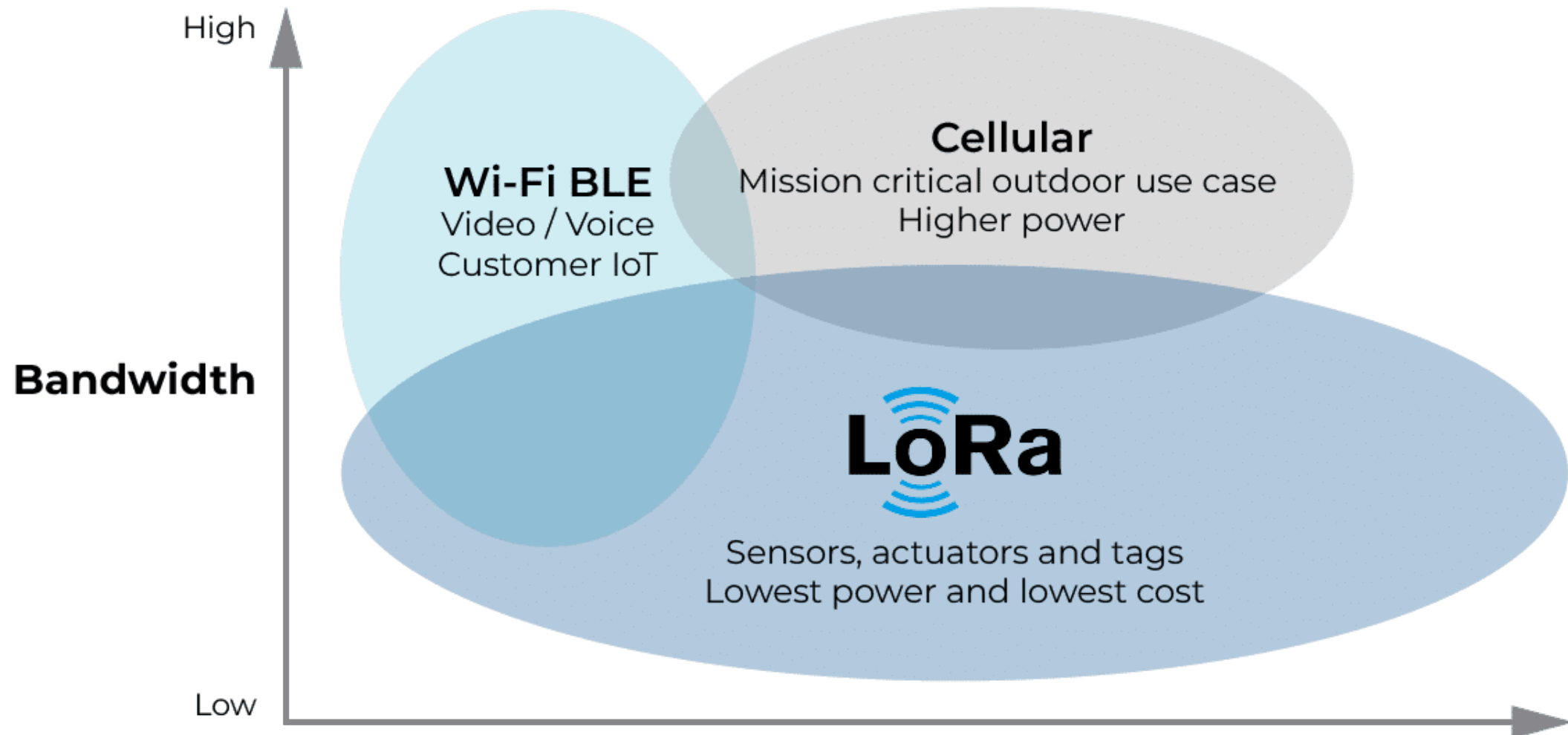
Section 706 of the Communications Act of 1934 gives the president authority to suspend telecommunications during an emergency (constitutionality not clear).

No direct government kill switch, but internet companies are largely American so president can exert political pressure.

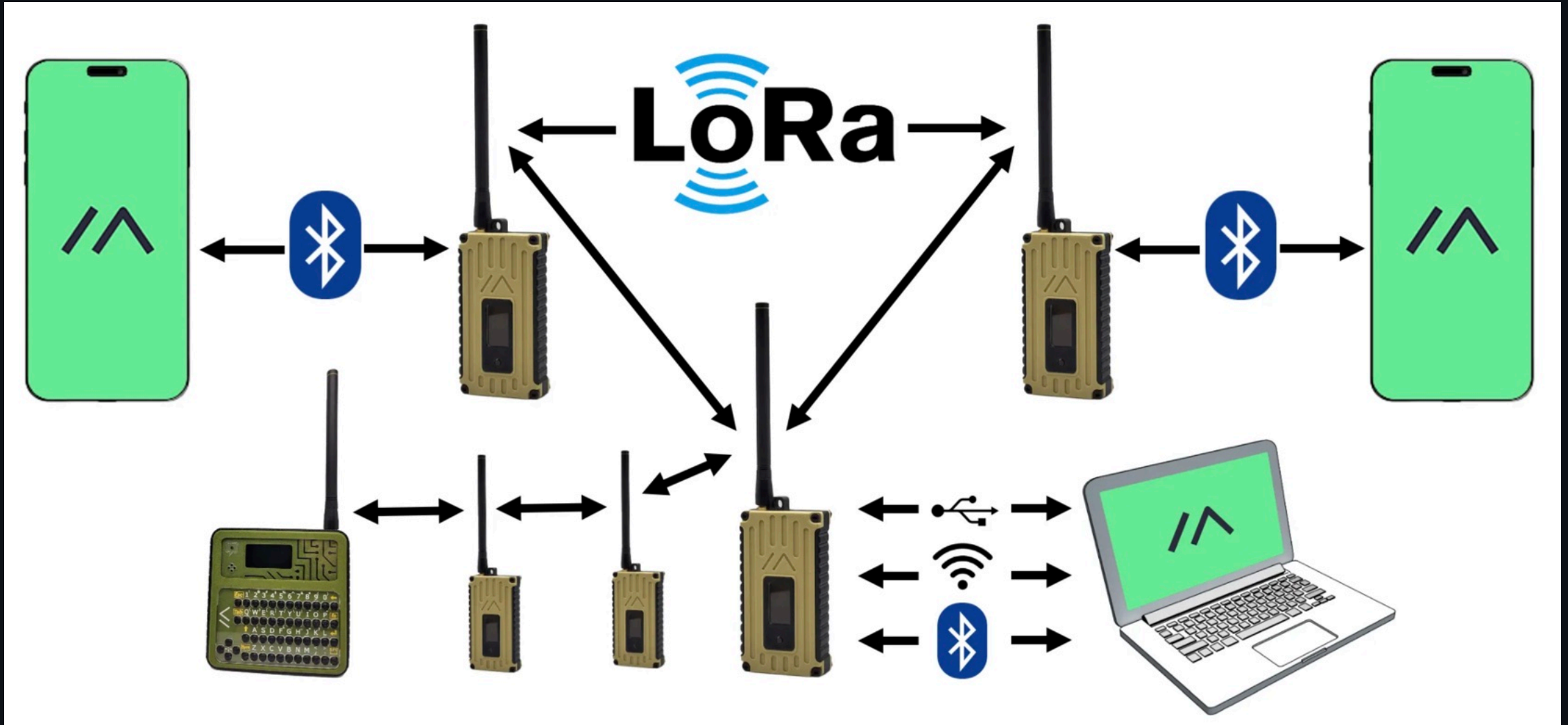
Coordination Matters



LoRa: No licenses, many devices



Meshtastic.org

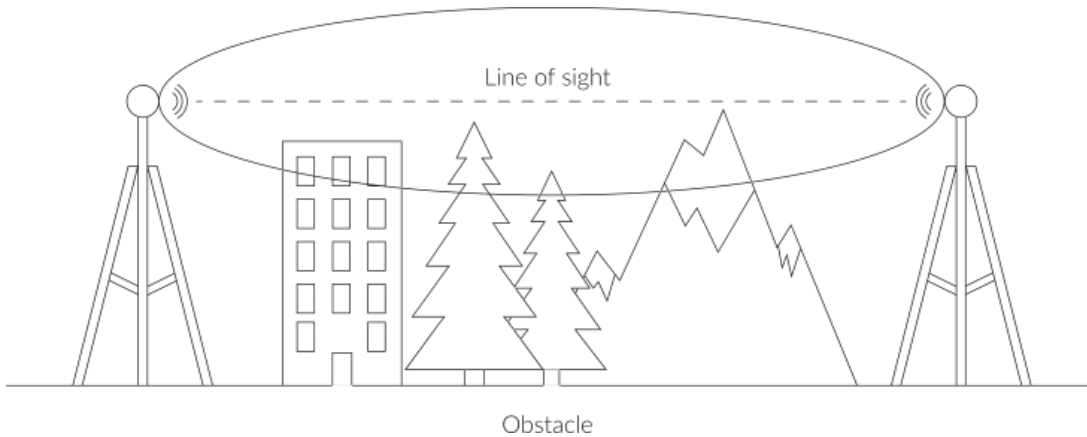


Meshtastic

Creates a mesh network.

Can either be on the public Meshtastic network or can generate security keys.

Range



Rural: 10-20 km

City: 1-10 km

As much as 300+ km claimed!

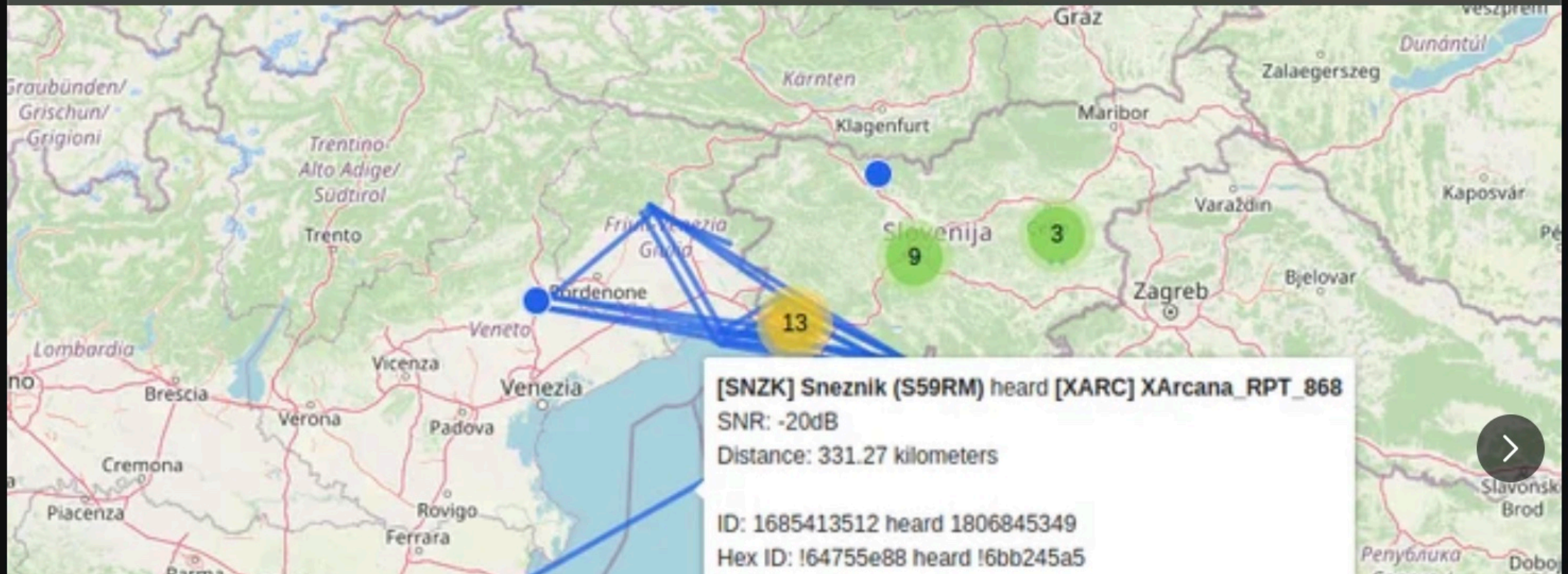


r/meshtastic • 1y ago

Martin_R7



Mountain to mountain 331 km



So what's it going to cost?

More affordable out of the box option

Wismesh Pocket 2: \$80 each

Bluetooth connectivity to devices

Mobile

Plug and Play



Raspberry Pi DIY Option

Not really any cheaper but
much greater range.

Requires some technical work.



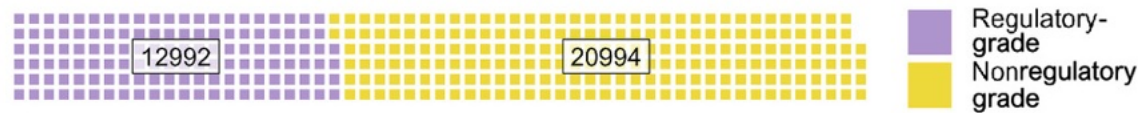
Price Breakdown - DIY Option

Component	Product	Price
Microcontroller	Raspberry Pi Zero 2 W	\$15
LoRa HAT	Adafruit RFM9x LoRa HAT (915MHz/868MHz)	\$25-35
Alternative HAT	Elecrow LoRa RFM95 IoT HAT	\$25-35
MicroSD Card	SanDisk Ultra 32GB Class 10 A1	\$8-12
Power Supply	Official Raspberry Pi 5V 2.5A Micro USB PSU	\$8-10
Antenna	Upgraded 915MHz/868MHz 5dBi LoRa Antenna with SMA connector	\$12-18
Case	Raspberry Pi Zero case (standard)	\$5-8
Cables/Adapters	Mini HDMI to HDMI adapter (for setup)	\$6-8
TOTAL		\$78-98

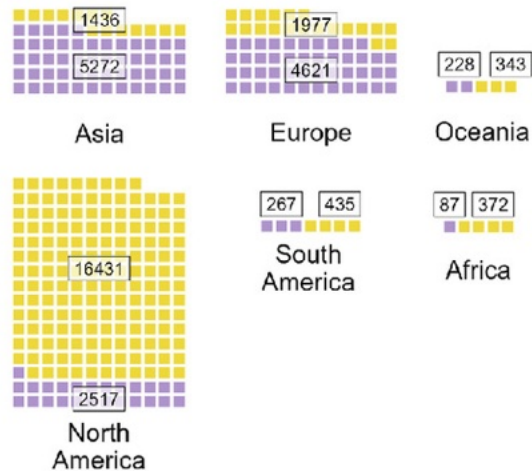
Persistent Inequalities in Air Quality Monitoring

i) Number of total monitoring stations

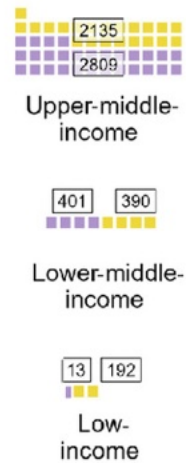
A Global



B Split by region

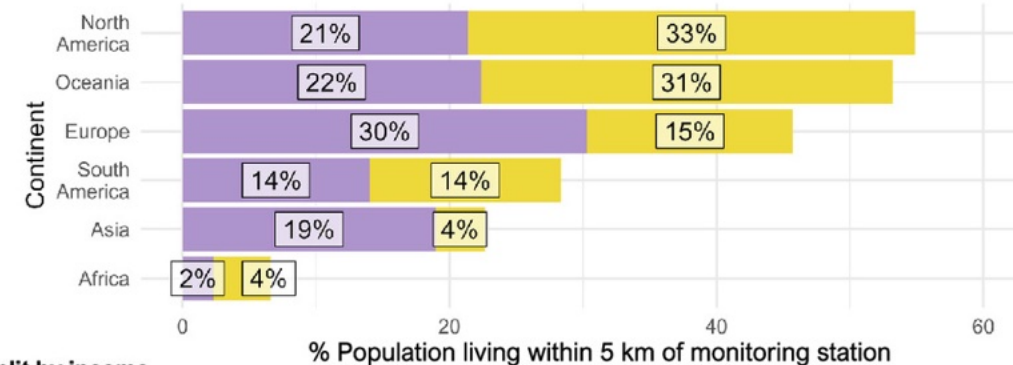


C Split by income

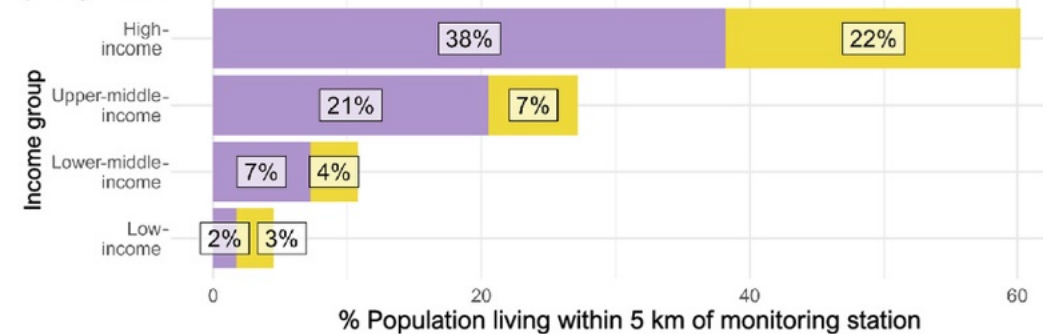


ii) Population living within 5 km of monitoring station

D Split by region

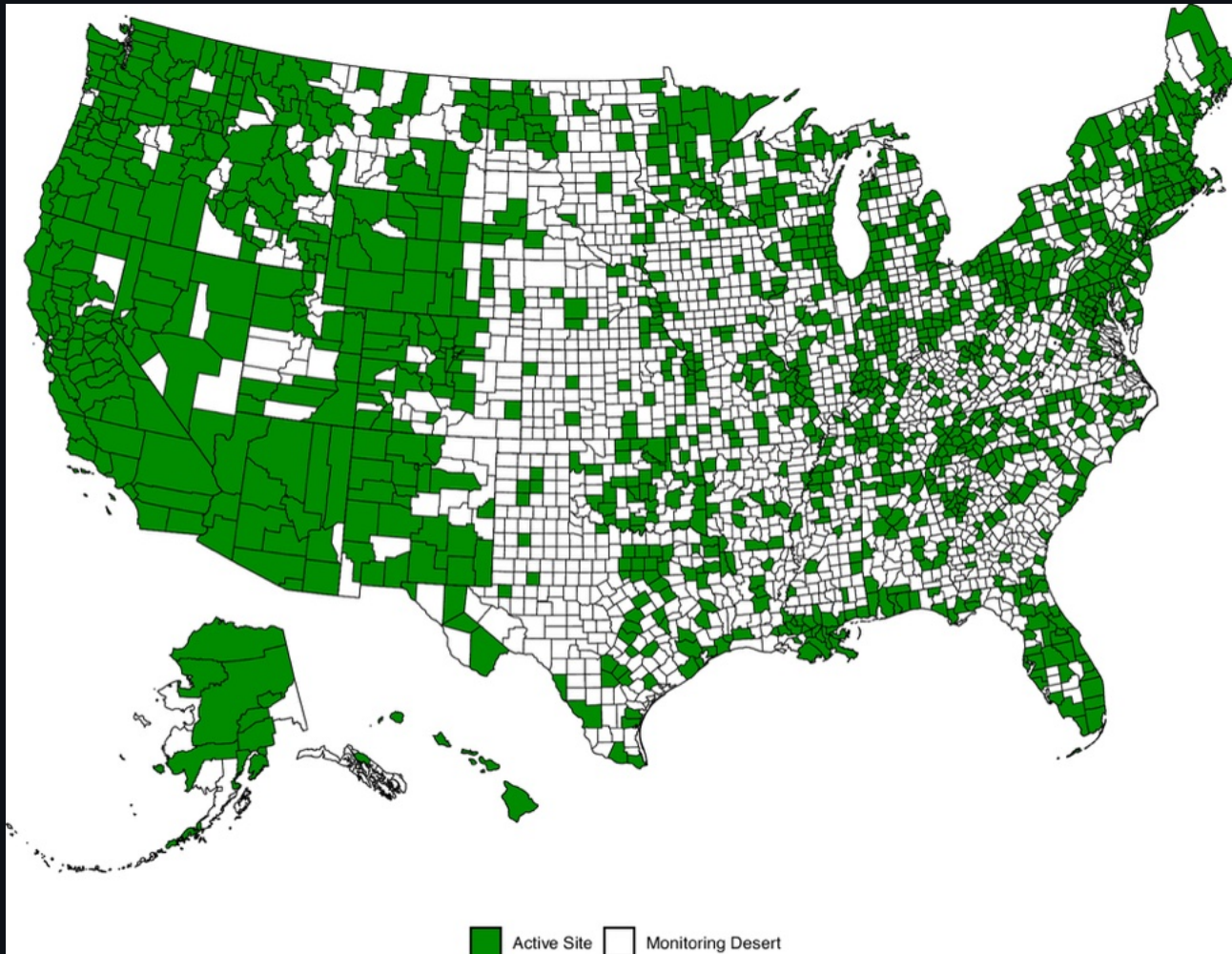


E Split by income



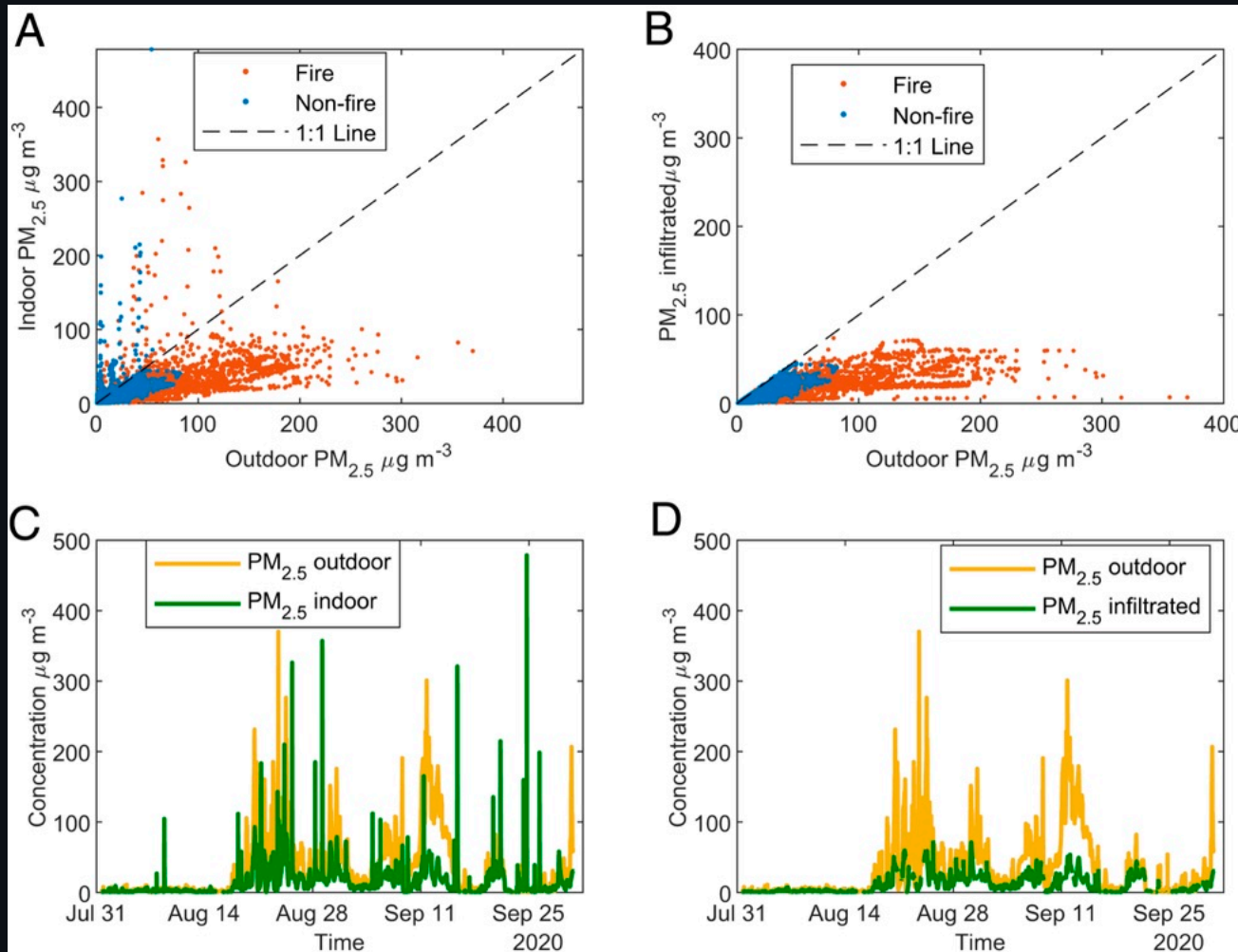
<https://doi.org/10.1073/pnas.2423259122>

Monitoring Deserts in the US

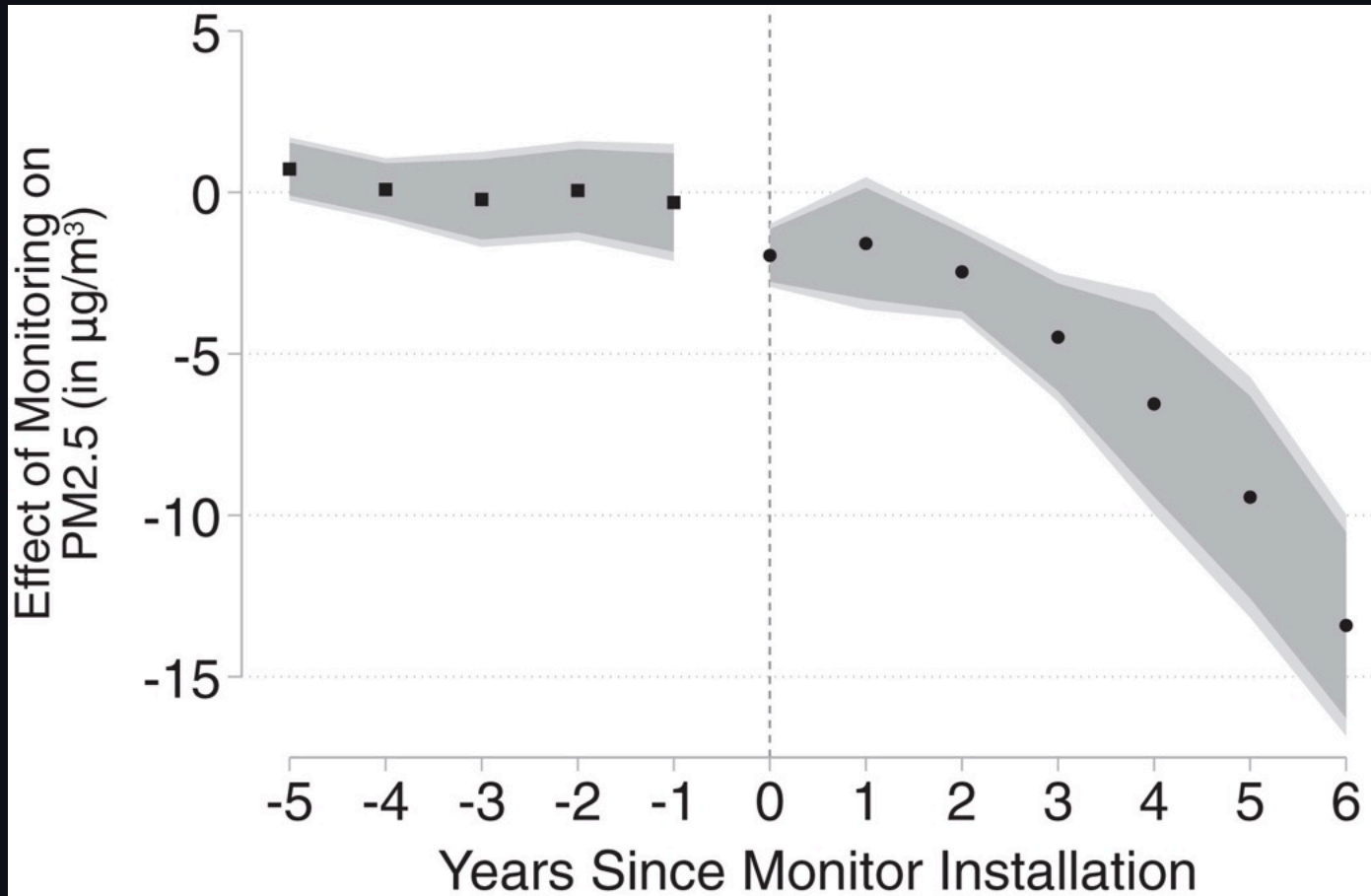


<https://doi.org/10.1073/pnas.2425310122>

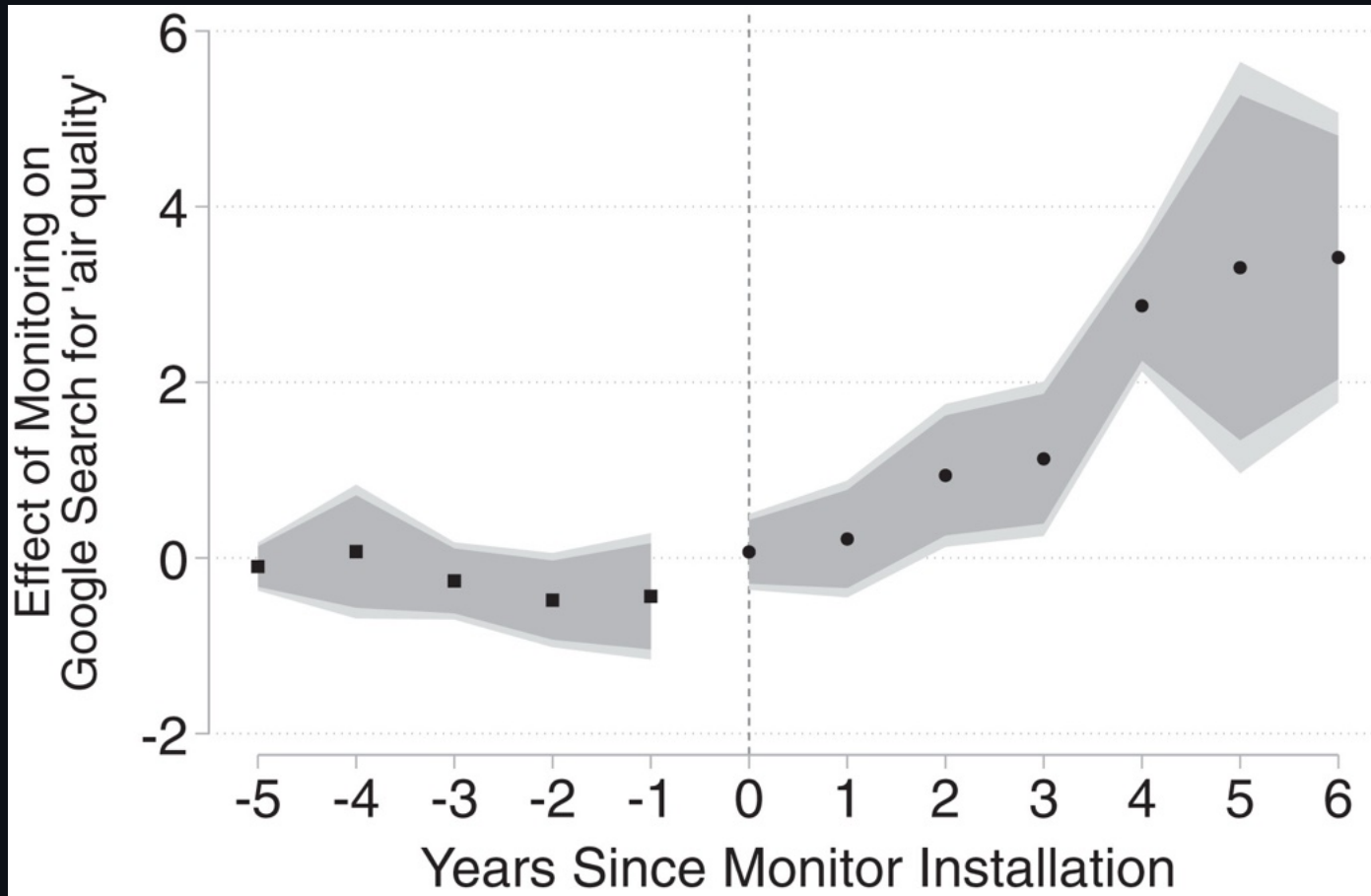
Bad Outdoor Air Quality = Bad Indoor Air Quality



US Embassy Tweets about Bad Air Quality: Local Governments Act



It's all about getting attention



<https://doi.org/10.1073/pnas.2201092119>

Risk of Air Quality Impacts: Low Cost Sensors Enabled This

RACE/ETHNICITY VARIABLES							
	Industry: SO ₂	Energy: SO ₂	Energy: NO _x	Agriculture: NH ₃	Transport: NO _x	Residential: OC	Commercial: NO _x
% White	Negative Association	No association	Negative Association	No association	No association	No association	Positive/Negative
% Black	Negative Association	No association	Negative Association	Positive Association	No association	Negative Association	Positive Association
% Asian	No association	No association	Positive Association	No association	Negative Association	Positive/Negative	Positive Association
% American Indian	Positive Association	No association	Positive Association	Positive Association	No association	Positive Association	No association
% Hispanic	Positive Association	No association	Positive Association	Positive Association	No association	Negative Association	No association
SOCIOECONOMIC VARIABLES							
% Unemployment	Positive Association	No association	No association	Negative Association	Negative Association	No association	Positive/Negative
% Poverty	Positive Association	Positive Association	Positive/Negative	Negative Association	Negative Association	Positive/Negative	Negative Association
Median Family Income	Negative Association	Negative Association	Negative Association	Positive Association	Positive/Negative	Negative Association	Positive/Negative
Median Property Value	No association	Negative Association	No association	Positive Association	Positive Association	Positive/Negative	Negative Association
Positive Association Negative Association No association Positive/Negative							

<https://doi.org/10.1038/s41467-023-43492-9>

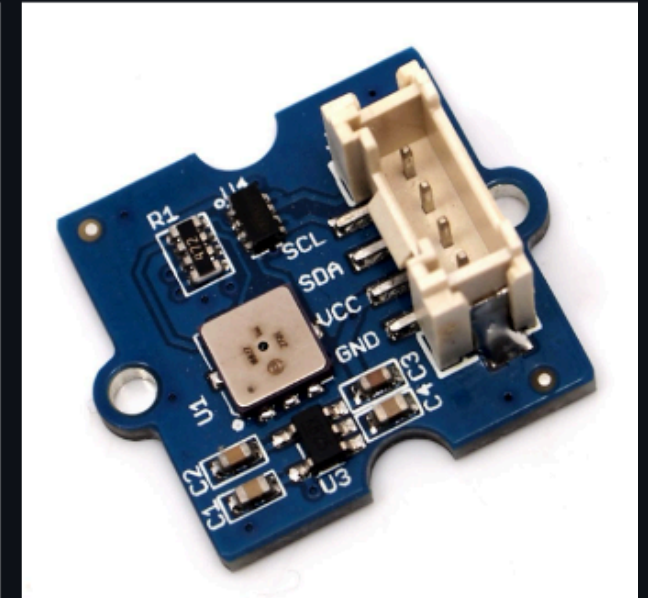
Air Alert: UM, Tribes Join Forces to Protect Air Quality

March 19, 2025



<https://www.umt.edu/news/2025/03/031925purp.php>

Sensors Being Used



Dust (PM 2.5)

Barometric Pressure/Humidity

Volatile Organic Compounds (VOCs)

Want one?

Email me at jakehosen@pm.me

Effects on Public Health

Air pollution, a preventable risk

SLCPs, particularly O_3 and BC and co-pollutants, which are important parts of PM2.5 air pollution, are harmful to human health. Globally, PM2.5 is the leading environmental cause of poor health and premature death.

DISEASES DUE TO:

● PM2.5 AIR POLLUTION

● O_3



- Heart attacks
- Strokes, heart disease
- Congestive heart failure



- Lung cancer
- Chronic bronchitis
- Asthma
- Emphysema
- Scarred lung tissue



- Low birth weight

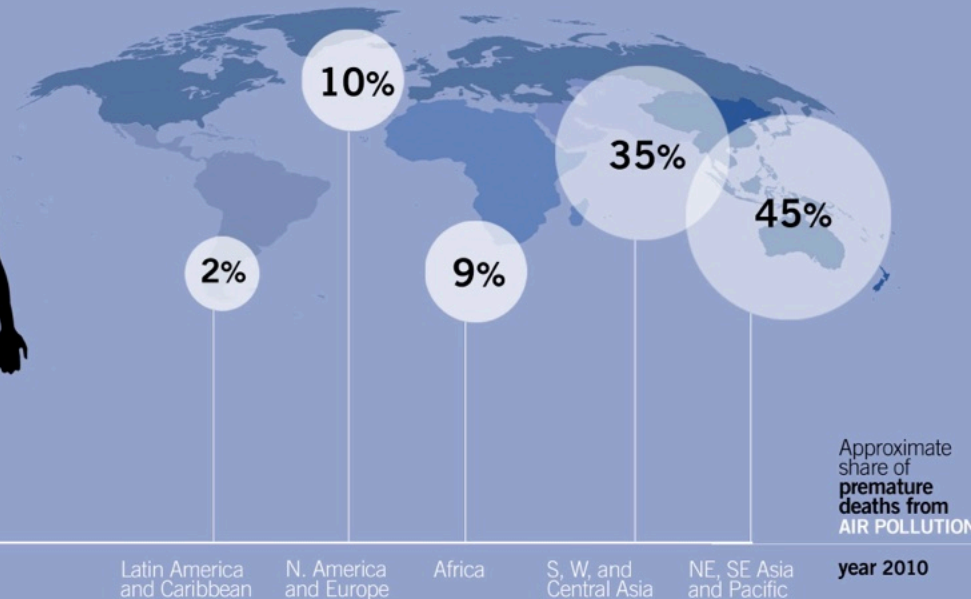


Globally, **air pollution** is the **2nd leading risk factor** for the global burden of disease in 2010, behind high blood pressure, and together with **tobacco smoking**, including second hand smoke.

PREMATURE DEATHS YEAR 2010

GLOBALLY, **AIR POLLUTION** IS RESPONSIBLE FOR:

- 3,500,000 From **indoor** PM2.5 pollution
- 3,200,000 From **outdoor** PM2.5 pollution
- 150,000 From **ozone** pollution



Volatile Organic Compounds (VOCs)

- There are a lot:
 - Ethanol
 - Methane (Greenhouse Gas)
 - Benzene (Air Pollution)
 - Geosmin (Natural Compound Released by Soil Bacteria)
 - Terpenes (Compounds released by plants)
 - Many human sources (paint, sprays, formaldehyde, peroxide)
- Can react with other air pollution to form ground-level ozone.

